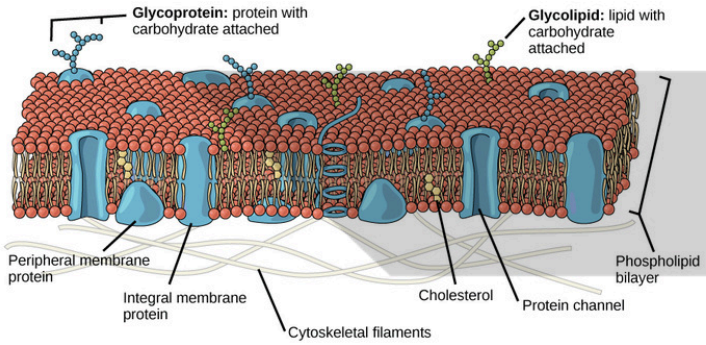
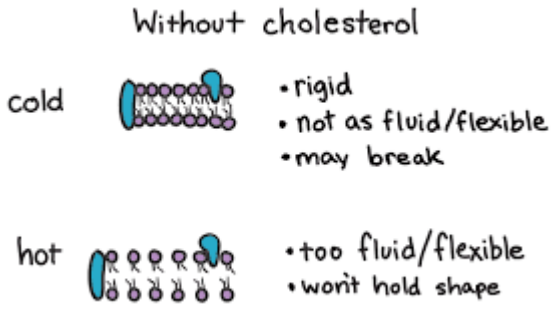
1. Cell Membrane and Transport

1.1. Fluid Mosaic Membranes

- Fluid Mosaic Model:
  - Individual phospholipids and some protein molecules move around (by diffusion); hence the word 'fluid' is used
  - The word 'mosaic' describes the pattern produced by scattered protein molecules when the surface of the membrane is viewed from above



- Phospholipid Bilayer:
  - In aqueous environments, the hydrophobic region (fatty acid tails) of the phospholipid molecules face inwards and away from the water, whereas the hydrophilic region (phosphate heads) face outwards and towards the water
  - This causes the phospholipid molecules to arrange into a 'bilayer.'
  - This provides the basic structure of membranes
  - It is selectively permeable
  - It acts as a barrier to most water-soluble substances
- Fluidity of the phospholipid bilayer:
  - The more unsaturated the tails, the more fluid the membrane as unsaturated fatty acid tails are bent because of the kinks due to double bonds and, therefore, fit together more loosely
  - The longer the tail, the less fluid the membrane
  - The higher the temperature, the more the kinetic energy of the phospholipid molecules and the higher the fluidity
  - Cholesterol:
    - Regulates the fluidity of membranes
    - At low temperatures, cholesterol increases the membrane's fluidity, preventing it from becoming too rigid.
    - At higher temperatures, it helps stabilize cells when the membrane could otherwise become too fluid.



- Other roles of cholesterol:
  - Helps with mechanical stability
  - Its hydrophobic region prevents polar molecules from passing through the membrane, e.g. in the myelin sheath.

1.2. Components of the Phospholipid Bilayer

- Glycolipids and Glycoproteins: Carbohydrate chains that are attached to membrane protein (glycoprotein) and phospholipids (glycolipid) project out into the watery fluids surrounding the cell, where they form hydrogen bonds to stabilize the membrane structure.
  - Carbohydrate chains act as receptors, mainly:
    - Signalling Receptors: The receptors recognise messenger molecules like hormones and neurotransmitters. When the messenger molecule binds to the receptor, chemical reactions are triggered inside the cell.
    - Endocytosis: This group of receptors binds to molecules to be engulfed by the cell surface membrane.
    - Cell Adhesion: binding cells to other cells in tissues and organs. Some glycolipids and glycoproteins act as antigens, allowing cell-cell recognition.
- Proteins: Transport proteins provide hydrophilic channels for ions and polar molecules. Enzymes catalyse the hydrolysis of molecules. Cytoskeletons made of protein filaments help maintain the shape of the cell.
- Channel Proteins:
  - Water-filled pores that allow charged substances, usually ions, to diffuse through the membrane
  - They have a fixed shape and can be gated to control ion exchange
  - This does not use ATP and is in facilitated diffusion

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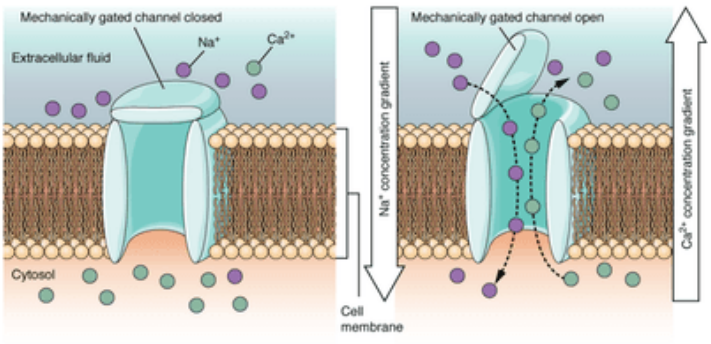
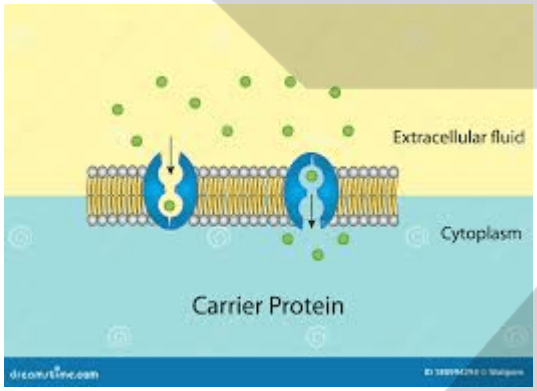


- Carrier Proteins:
  - It can flip between two shapes and is mainly in active transport, where it uses ATP to change shape and carry ions/molecules up the concentration gradient.
  - It is also involved in passive transport (facilitated diffusion) down the concentration gradient without the use of energy.



- Cell Surface Receptors:
  - These are present in membranes and bind with particular substances, e.g., hormones, which are chemical messengers which circulate in the blood but only bind to specific target cells
- Cell Surface Antigens:
  - These act as cell identity markers
  - Each type of cell has its antigen
  - This enables cells to recognise other cells and behave in an organized way

1.3. Cell Signalling

- Cell Signalling:
  - The molecular mechanisms by which cells detect and respond to external stimuli, including communication between cells
    - Specific chemicals called 'ligands' are secreted by cells
    - These ligands are transported to specific target cells in the bloodstream
    - The ligand binds to the cell surface receptor on the target cell, which is complementary in shape and specific to that particular ligand
    - This binding about a conformational change in the shape of the receptor, and the message is passed to the inside of the cell (signal transduction)
    - Changing the shape of the receptor allows it to interact with the G protein, which brings about the release of a 'second messenger' (a small molecule which diffuses through the cell relaying the message)
    - The second messenger activates a cascade of enzyme-catalysed reactions, which brings about the required change
    - This is an active process involving ATP use

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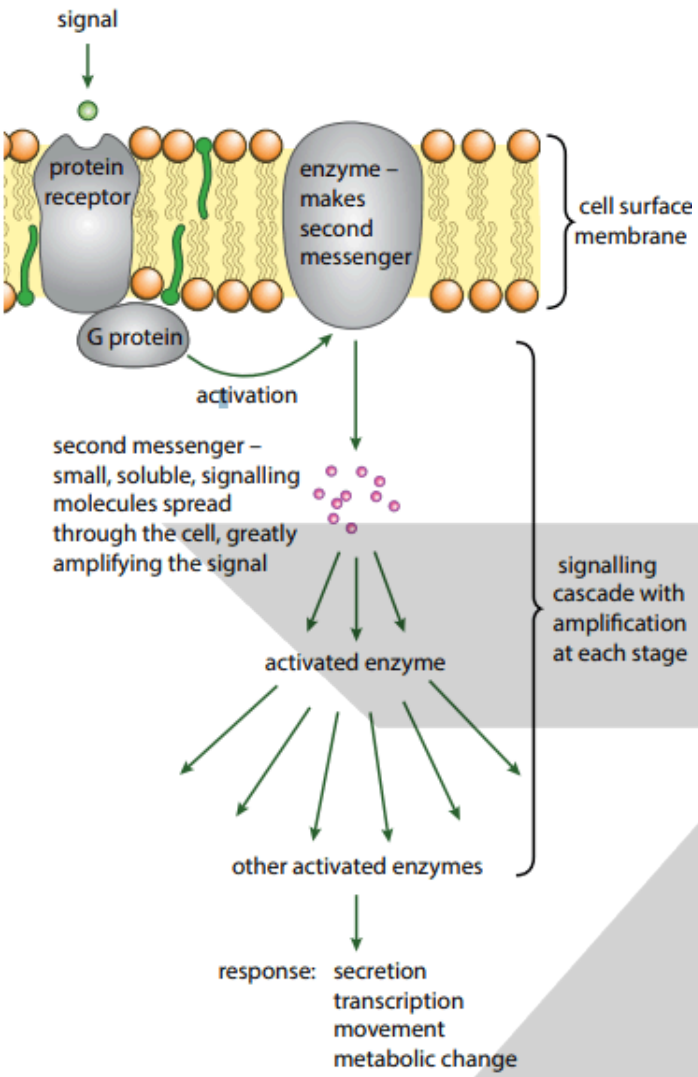
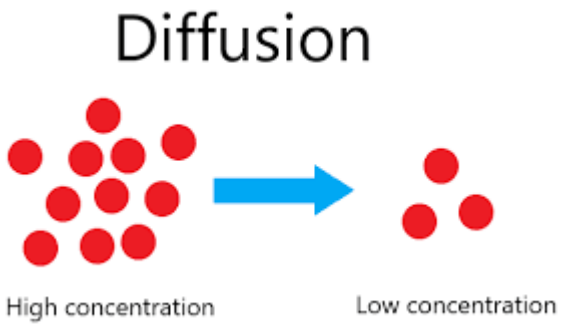
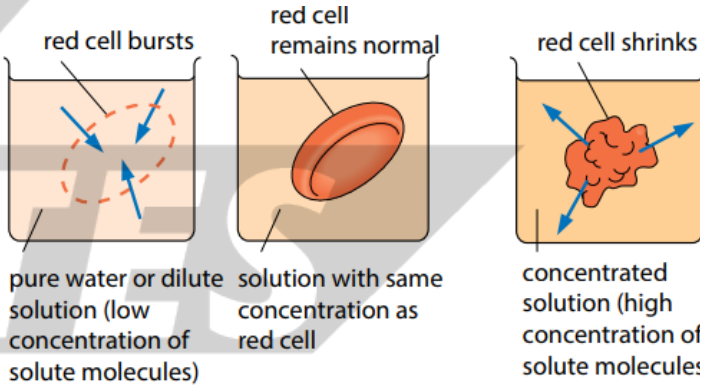


1.4. Movement Into and Out of Cells

- Diffusion:
  - The net movement of molecules or ions from a region of high concentration to a region of low concentration
  - It is a passive process (molecules have natural kinetic energy)
  - As a result of diffusion, molecules reach equilibrium
    - A **steeper concentration gradient, higher temperature** and **increased surface area** all increase the rate of diffusion
    - Nonpolar molecules can pass directly through the membrane, e.g., steroids, hormones
    - Gases (like oxygen and carbon dioxide) can diffuse through the membrane directly
    - Despite being polar, water can diffuse through directly as it is a small molecule



- Facilitated Diffusion:
  - Movement of molecules from a region of high concentration to a region of low concentration down a concentration gradient (passive movement)
  - Molecules go through transport proteins instead of passing directly through phospholipids
  - This allows for the passage of large polar ions and molecules, e.g., glucose, amino acids, Na+, Cl-
- Osmosis:
  - The diffusion of water molecules from a region of higher water potential (ψ) (less negative) to a region of lower ψ (more negative) through a selectively permeable membrane
    - ψ is the tendency of water to move out of a solution
    - Pure water has 0 ψ
    - Negative ψ means that the solution has more solute than the solvent
- In red blood cells (lacking cell wall):



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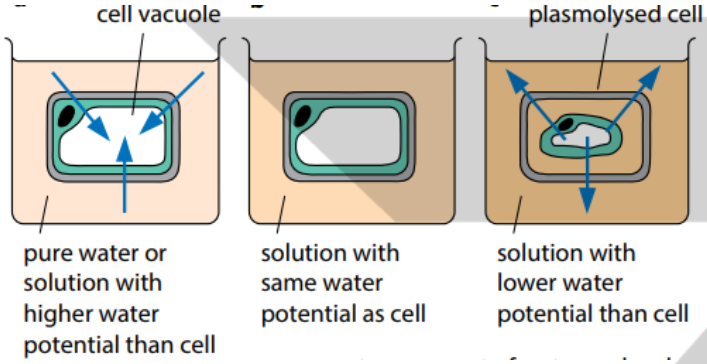
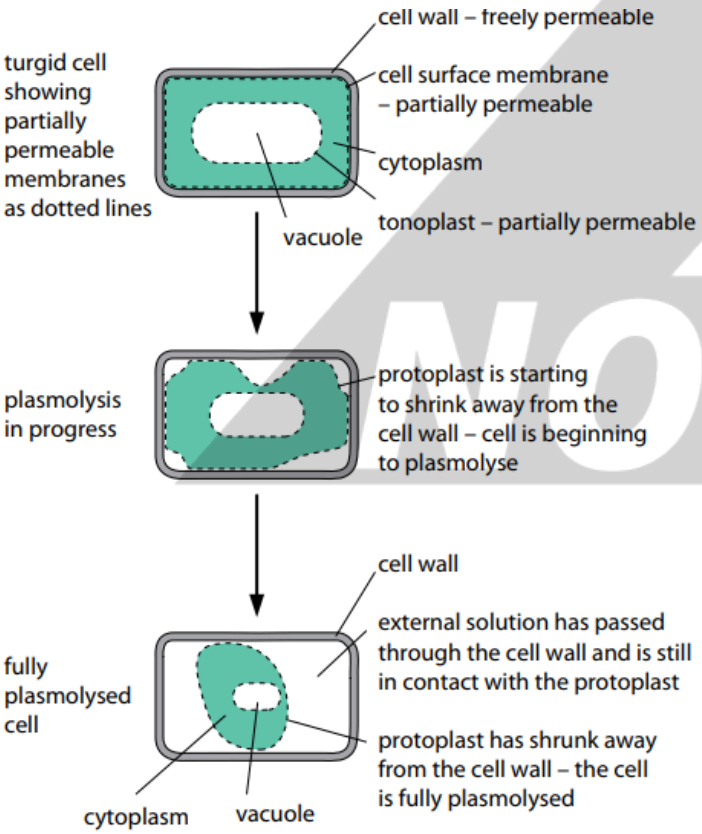
- In plant cells (having cell walls):
  - **Protoplast:** the living part of the cell inside the cell wall
  - In pure water:
    - Water enters the cell by osmosis, and the cell wall pushes back against the expanding protoplast, building up pressure rapidly, becoming turgid
  - In a concentrated solution:
    - Water will leave the cell by osmosis
    - The protoplast gradually shrinks until it exerts no pressure on the cell wall
    - The protoplast continues to shrink and pulls away from the cell wall, so the cell is plasmolysed



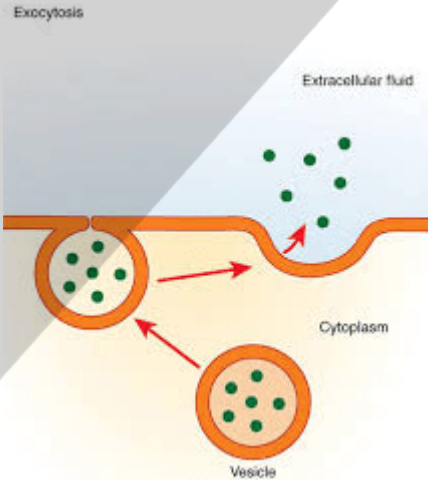
- How plasmolysis occurs:



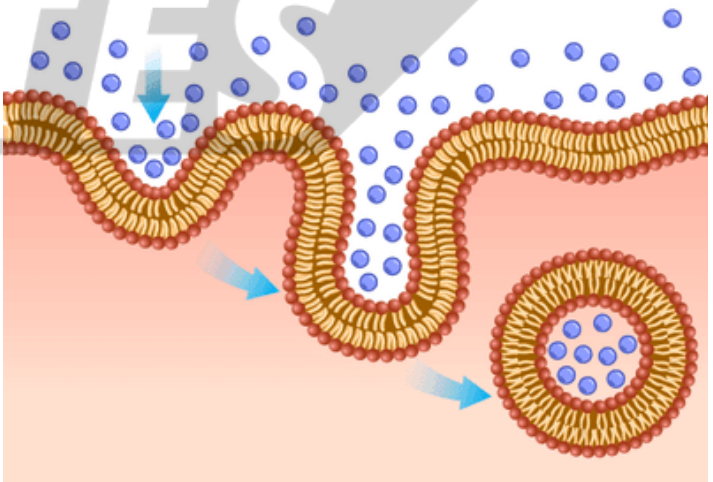
- Active Transport:
  - Movement of substances from a region of low concentration to a region of high concentration against a concentration gradient
  - This occurs via specific carrier proteins for specific ions/molecules that use energy from ATP

1.5. Bulk Transport

- Exocytosis
  - This is the movement of substances out of the cell
  - A secretory vesicle from the Golgi body moves towards the plasma membrane with the help of the cytoskeleton, using energy from ATP
  - The vesicles fuse with the cell surface membrane, releasing the contents outside



- Endocytosis
  - Involves the engulfing of the material by fusing with the plasma membrane to form an endocytic vacuole in the form of phagocytosis (bulk uptake of solids) or pinocytosis (bulk uptake of liquids) using ATP



1.6. Investigations

Demonstrating diffusion using Visking tubing

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- Visking tubing is a partially permeable, non-living membrane made from cellulose.
- It has pores which can allow small molecules like glucose to diffuse through but won't allow larger molecules like starch to diffuse through
- Fill a length of visking tubing with a mixture of starch and glucose
- Suspend the tubing in water for a period of time
- Carry out Benedict's and Iodine tests on the water at suitable time intervals to check for the presence of glucose and starch.
- A positive result for Benedict's test but a negative result for the iodine test will be obtained.

Demonstrating diffusion using plant tissue

- Pieces of beetroot can be placed into water at different temperatures
- Any damage to the cell membranes results in the red pigment leaking out of cells by diffusion
- What you are seeing is the diffusion of the red dye from a region of high concentration in the beetroot vacuoles to a region of low concentration in the solution outside the beetroot
- Only at temperatures at which the cell membrane starts to damage does the red dye start to diffuse out (diffusion is normally prevented by the partially permeable nature of the cell membranes)

Investigating the effect of size on diffusion

- Surface area to volume ratios decrease with increasing size of 3D structures.
- Prepare agar blocks of different SA: Volume ratios
- Place the blocks in dilute hydrochloric acid
- Measure the time taken for the blocks to turn pink
- The SA: Volume ratio that takes the least time to give colour change indicates the fastest diffusion

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